

The question asked in 8 out of 10 machine learning interviews

Why does Lasso regression create sparsity, while Ridge regression does not?

A mathematical breakdown of feature selection using single-variable calculus.

Ridge: Shrinks values toward zero.

Feature	Coefficient
Feature A	0.004
Feature B	0.001
Feature C	-0.003



Lasso: Forces values to exactly zero.

Feature	Coefficient
Feature A	0.000
Feature B	0.000
Feature C	0.000

Starting with the baseline: Single-variable linear regression

Before adding any regularisation, calculating the coefficient (m) requires dividing the variance components. We will use Y and X^2 as visual shorthand for these terms.

The diagram shows the formula for the coefficient m in single-variable linear regression. The formula is presented as a fraction. The numerator, $\sum (y_i - \bar{y})(x_i - \bar{x})$, is enclosed in a blue rounded rectangle. A blue arrow points from the label "Numerator (Y)" to this rectangle. The denominator, $\sum (x_i - \bar{x})^2$, is enclosed in a grey rounded rectangle. A grey arrow points from the label "Denominator (X^2)" to this rectangle. The entire formula is preceded by $m =$.

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

Numerator (Y)

Denominator (X^2)

The absolute value in the Lasso loss function breaks standard calculus

To find the optimal m , we must differentiate this loss function and set it to zero. However, the absolute value function $|m|$ has a sharp corner at zero, making it mathematically non-differentiable at that exact point.

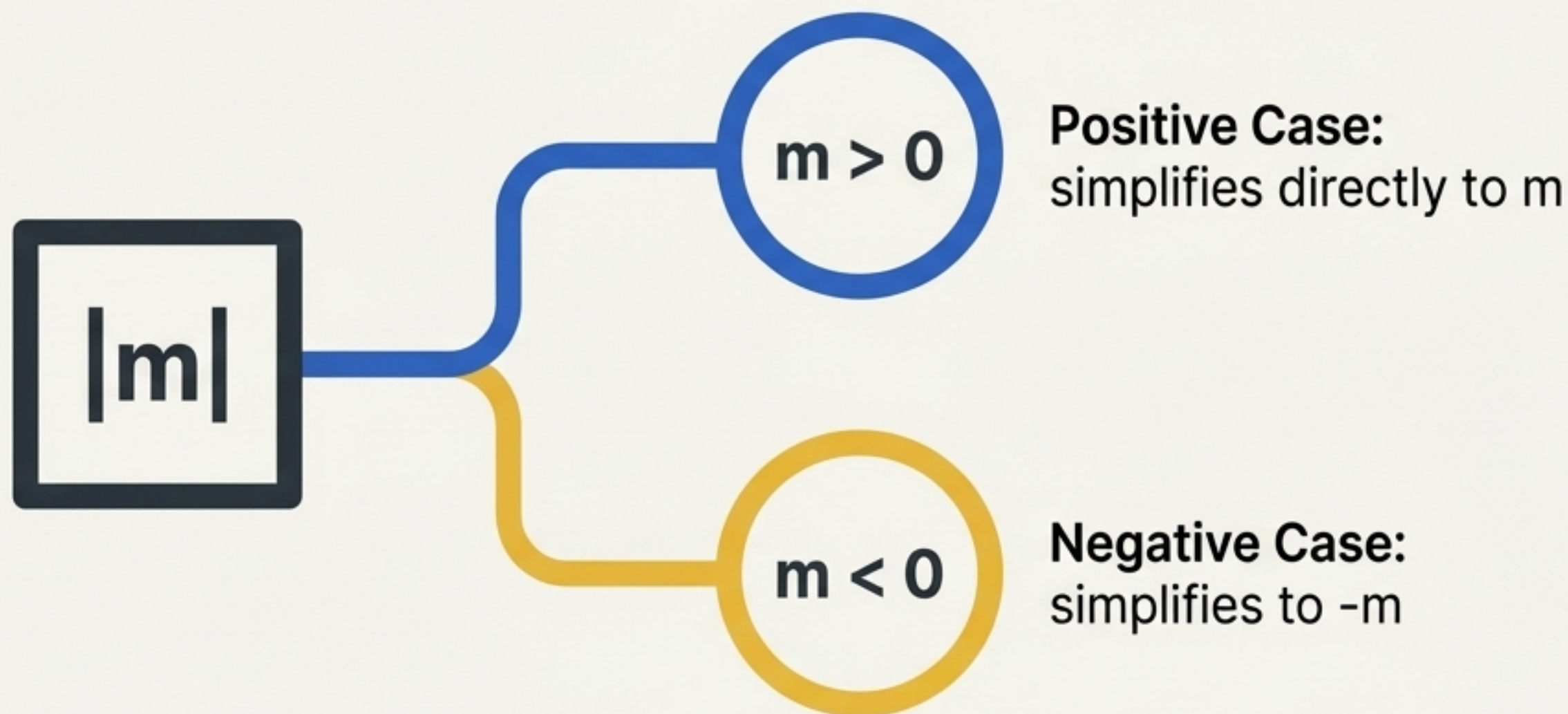
$$\sum (y_i - (mx_i + b))^2 + \lambda |m|$$

Non-differentiable
corner



Solving the absolute value problem requires two separate cases

Because we cannot differentiate exactly at zero, we must calculate the derivative for positive coefficients and negative coefficients independently to see how the maths behaves.



Case 1: For positive coefficients, Lasso subtracts directly from the numerator

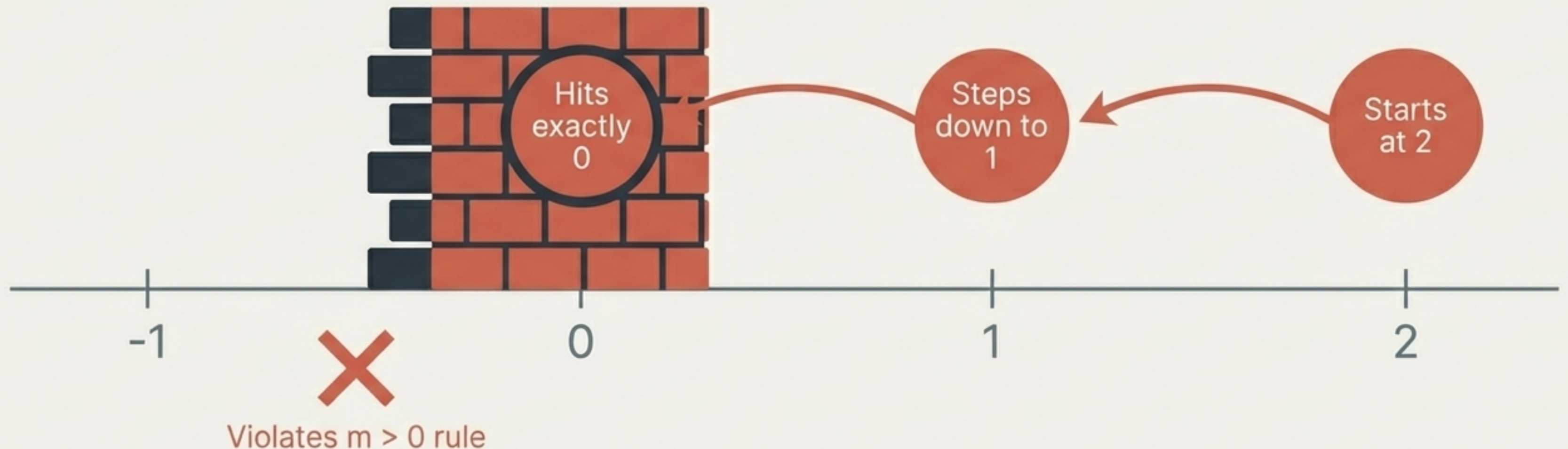
When we differentiate the $m > 0$ case and isolate m , the penalty term (λ) manifests as a direct subtraction from our baseline numerator.

$$m_{\text{lasso}} = \frac{\text{Numerator (Y)} - \lambda}{\text{Denominator (X}^2\text{)}}$$

Penalty actively subtracts from the top half

The self-correction mechanism: Why positive values get stuck at zero

As the penalty grows, the value marches toward zero. If the subtraction attempts to push the coefficient into negative territory, the mathematical condition ($m > 0$) breaks, trapping the value exactly at zero.



Case 2: For negative coefficients, Lasso adds directly to the numerator

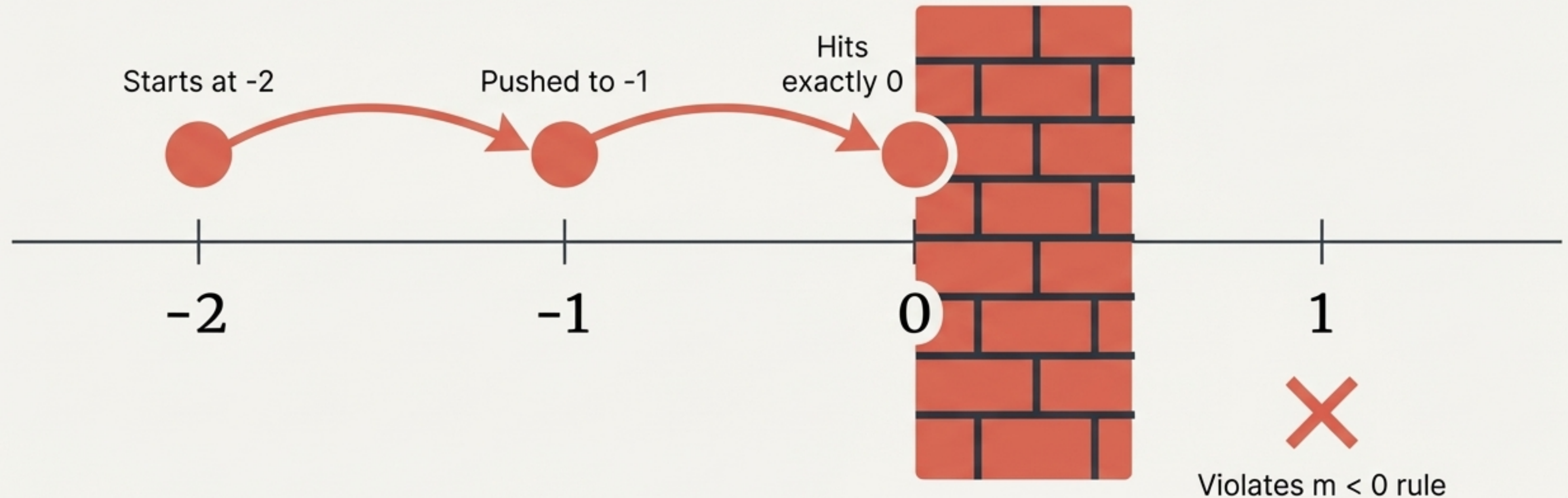
When calculating for $m < 0$, the mathematical signs flip. The penalty term now acts as a direct addition to the negative baseline numerator.

$$m_{\text{lasso}} = \frac{\text{Numerator } (-Y) + \lambda}{\text{Denominator } (X^2)}$$

Penalty actively adds to the negative top half

Negative coefficients are pushed up to **zero**, and halted

The algorithm actively forces negative coefficients upwards. Once they hit zero, crossing the threshold would trigger the opposite formula. The value is mathematically deadlocked at zero.



Ridge regression operates in a completely different mathematical space

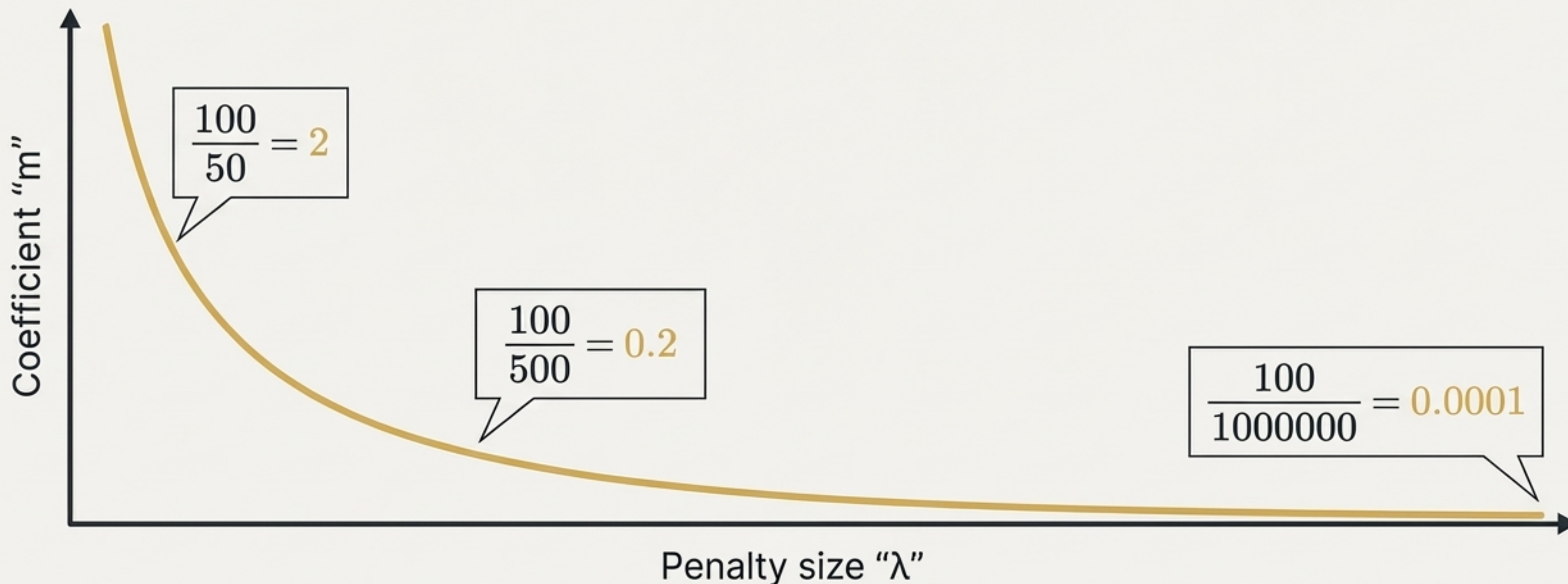
Unlike Lasso's L1 absolute value penalty, Ridge uses an L2 squared penalty (λm^2). When differentiated, this places the λ penalty exclusively in the denominator.

$$m_{\text{ridge}} = \frac{\text{Numerator } (Y)}{\text{Denominator } (X^2) + \lambda}$$


Penalty is trapped in the denominator

Denominator dynamics: Asymptotically small, but never zero

No matter how unimaginably large the penalty (λ) becomes in the denominator, you are simply dividing the baseline by a larger number. The coefficient will become microscopic, but unless the numerator is naturally zero, the result cannot logically equal zero.



The definitive mechanical contrast

The fundamental difference lies entirely in fractional placement.

Lasso

$$-\lambda$$



Can equal exactly 0

Numerator Subtraction. Actively drives values to a hard zero.

Ridge

$$\frac{\quad}{+ \lambda}$$



Cannot equal exactly 0

Denominator Scaling. Asymptotically shrinks values without deleting them.

The 30-second interview answer

Lasso regression creates sparsity because its **L1 penalty** places the **λ term** in the **numerator** of the weight update formula, allowing it to **subtract or add** the coefficient to **exactly zero**. If the **penalty tries to push the value past zero**, the **gradient's sign flips**, mathematically **trapping** the coefficient at zero.

Ridge regression fails to create sparsity because its **L2 penalty** places **λ** in the **denominator**, meaning the coefficient is only ever **divided by a larger number**, **shrinking asymptotically** but **never reaching true zero**.